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Recent progress in copper-catalyzed electrophilic amination

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Copper-catalyzed coupling reactions have been recognized as one of the most useful strategies for the formation of C–N bonds. This perspective gives an overview of the recent developments in copper-catalyzed electrophilic amination for the construction of various amines and their derivatives, including the electrophilic amination of various organometallic reagents and direct C–H bonds as well as the annulative electrophilic amination of *o*-alkynylphenols and *o*-alkynylanilines.

1. Introduction

Amines are privileged structural motifs in many biologically active compounds, pharmaceutical targets, and functional materials.¹ Metal-promoted C–N cross-coupling reactions provide a powerful and convergent approach towards amines and their derivatives. Recently, Pd- or Cu-catalyzed processes from nucleophilic amines with electrophilic aryl halides (Buchwald–Hartwig C–N coupling reaction) have been extensively explored in this field (Scheme 1, route A).² With the Cu-catalyzed process now being the most powerful and reliable method for the synthesis of amines and their derivatives. Recent advances in metal-mediated C–H functionalization chemistry have enabled the direct oxidative C–H/N–H coupling

to be a complementary and potentially more effective process for the synthesis of amines (Scheme 1, route B).³ In addition, Cu-mediated oxidative amination of nucleophilic arylboronic acids with amines (Chan–Lam-type coupling) is a hallmark reaction in this field.⁴ On the other hand, an umpolung strategy – electrophilic amination using a reagent of type R₂N⁺ was considered as a viable alternative for the formation of the C–N bond and will remain an active area in constructing amines and their derivatives (Scheme 1, route C).

To date there are many reports on the formation of C–N bonds *via* electrophilic amination of carbanions and enolates with aminating reagents.⁵ Most of these reactions need active organometallic reagents such as RMgX or RLi, which are less substrate-tolerant. Due to the oxidizability of electrophilic nitrogen sources, direct reactions of organometallic reagents with aminating reagents often afford the products in modest yields. In general, Pd, Rh, Ru, Ir, Ni, Cu, and Fe complexes are the most investigated catalysts for these coupling reactions. Although different advantages could be specifically ascribed

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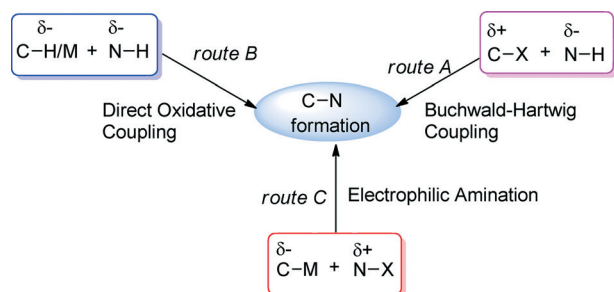
Xiaoyu Yan

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Xianghua Yang

Xianghua Yang obtained her masteral degree in 2007 from Prof. Chanjuan Xi's group, Tsinghua University. Then she joined the group of Prof. Heinz Berke at the University of Zurich where she became Doctor of Philosophy in 2012. Afterwards she spent one and a half years child-rearing until she took up the current position in the middle of 2013 as a "One-Hundred Young Talents" Associate Professor at Guangdong University of Technology. Her research interests include the methodology of organic chemistry, organometallics and organocatalysis.



Scheme 1 Construction of the C-N bond via cross-coupling reactions.

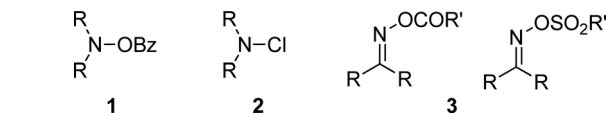
to each of these metals, Cu-catalysis represents a promising direction in the C-N formation reaction, due to its low cost, low toxicity, high efficiency and easy handling. In this perspective, we focus on recent advances in the field of copper-catalyzed electrophilic amination.

2. Electrophilic aminating reagents

In an electrophilic amination reaction, amino or substituted amino groups are transferred from certain aminating reagents to nucleophiles. Such reagents usually contain an electron-withdrawing functional group attached to nitrogen which would induce a partial positive charge at the nitrogen atom. Up to now, various types of aminating reagents such as *N*-haloamines, substituted hydroxylamines, oxaziridines, oximes, diazonium salts, azodicarboxylates, and azides have been reported to be effective. Three kinds of electrophilic aminating reagents, *O*-benzoylhydroxylamines **1**, *N*-chloroamines **2** and oxime esters **3** were most frequently used in recent copper-catalyzed electrophilic amination reactions and are discussed herein (Scheme 2).

2.1 *O*-Benzoylhydroxylamines

O-Benzoylhydroxylamines are the most widely used electrophilic aminating reagents amongst a number of *O*-substituted



Scheme 2 Typical electrophilic aminating reagents: *O*-benzoylhydroxylamines **1**, *N*-chloroamines **2**, and oxime esters **3**.

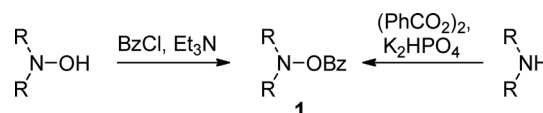
hydroxylamines. They can be easily prepared, either by benzoylation of hydroxylamines in the presence of a base (Scheme 3, left side)⁶ or *via* oxidation of secondary amines with benzoyl peroxide and an exogenous inorganic base in ethereal solvent as reported by Ganem and coworkers (Scheme 3, right side).⁷ Furthermore, Johnson and coworkers discovered that the oxidation of primary and secondary amines with benzoyl peroxide can afford *O*-benzoylhydroxylamines with high yields in a polar aprotic solvent at room temperature.⁶

2.2 *N*-Chloroamines

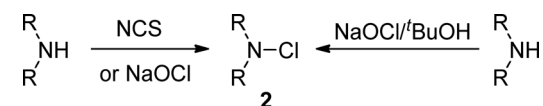
N-Chloroamine was one of the earliest investigated reagents for electrophilic amination. It can be prepared by treating the corresponding amines with NCS (*N*-chlorosuccinimide) or NaOCl (Scheme 4, left).⁸ Zhong *et al.* reported a modified protocol for the preparation of *N*-chloroamines *via in situ* generation of *tert*-butyl hypochlorite by NaOCl and *t*-BuOH (Scheme 4, right).⁹ In these procedures, the *N*-chloroamines can all be isolated simply by extraction.

2.3 Oxime esters

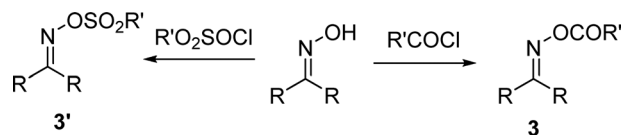
Oxime esters such as *O*-acyloximes and *O*-sulfonyloximes are a class of aminating reagents which have been widely used in electrophilic amination. The *O*-acyloximes and *O*-sulfonyloximes can be prepared by direct benzoylation or sulfonylation of oximes (Scheme 5),⁷ which have been obtained quantitatively *via* reaction of ketones or aldehydes with hydroxylamine hydrochloride.



Scheme 3 Synthesis of *O*-benzoylhydroxylamines.



Scheme 4 Synthesis of *N*-chloroamines.



Scheme 5 Synthesis of oxime esters.



Chanjuan Xi

Chanjuan Xi is a Professor of the Department of Chemistry, Tsinghua University. She graduated from Lanzhou University in 1986. She worked at the Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences (1986–1995). In 1996, she joined Hokkaido University and earned her Ph.D. in Pharmaceutical Sciences in 1999, and spent one year as a postdoctoral fellow in Catalysis Research Center, Hokkaido University (Prof. Tamotsu Takahashi). In 2000, she joined Tsinghua University as an Associate Professor and then she was promoted to a Professor in 2005. Her research interests are in organometallics and synthetic organic chemistry.

3. Electrophilic amination with various nucleophilic organometallic reagents

The electrophilic amination of carbanions, *i.e.*, the conversion of organometallic compounds to amines, has been known for a long time, and a large number of organometallic compounds have been successfully employed in electrophilic amination under catalyst-free or catalytic conditions. In this perspective, only the copper-catalyzed ones have been listed.

3.1 Electrophilic amination with Grignard reagents

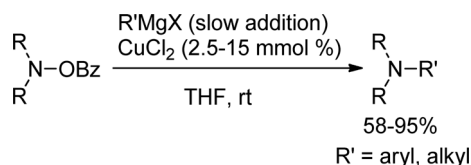
Grignard reagents (RMgX) can react with parent haloamines or alkylhydroxylamines to yield primary amines. This method has been broadly applied for the attachment of an amino group to organometallic compounds.^{5e} However, for reactions with *N*-substituted haloamines and hydroxylamines, the yields of the corresponding amines have been rather low due to the occurrence of side reactions such as halogenation and elimination.¹⁰ Johnson and coworkers reported a CuCl₂-catalyzed amination reaction of Grignard reagents with *O*-acylhydroxylamines (Scheme 6).¹¹ Both aryl and alkyl Grignard reagents were converted into the corresponding amines in good to high yields. In this amination process, the side reaction (*C*-acylation) was greatly decreased in the presence of a catalytic amount of CuCl₂.

The reaction of phenylmagnesium bromide with *O*-(2,4,6-trimethylphenylsulfonyl)oxime in a mixed solution of diethyl ether–toluene (1 : 4 by volume) at 75 °C gave aniline with 58% yield after 40 h. When the reaction was treated with 10 mol% of CuI, the reaction time decreased to 11 h without a significant change in the yield (Scheme 7).¹²

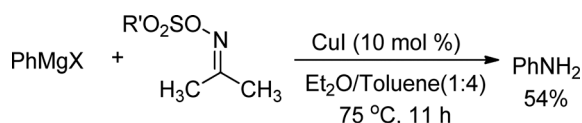
Narasaka and coworkers used benzophenone *O*-sulfonyloxime derivatives for the electrophilic amination of alkyl-, cycloalkyl-, and arylmagnesium reagents. 4,4'-Bis(trifluoromethyl)benzophenone *O*-methylsulfonyloxime was reacted with primary, secondary, tertiary, and cycloalkyl magnesium reagents in THF–HMPA solution at 0 °C in the presence of CuCN·2LiCl as the catalyst, providing the corresponding amines in high yields (Scheme 8).¹³

3.2 Electrophilic amination with organozinc reagents

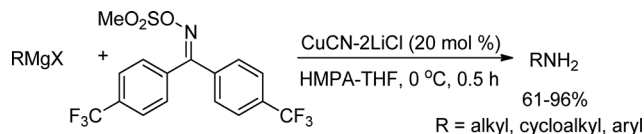
Organozinc compounds are an important class of main-group organometallic reagents. The bonds between carbon



Scheme 6 CuCl₂-catalyzed amination of Grignard reagents.



Scheme 7 CuI-catalyzed amination of Grignard reagents.



Scheme 8 CuCN·2LiCl-catalyzed amination of Grignard reagents.

and zinc have some covalent features, therefore, they display modest reactivities towards many electrophilic reagents. Such characteristics allow them to tolerate numerous functional groups;¹⁴ a number of electrophilic amination procedures have been reported in the literature that directly employ organozinc reagents.¹⁵

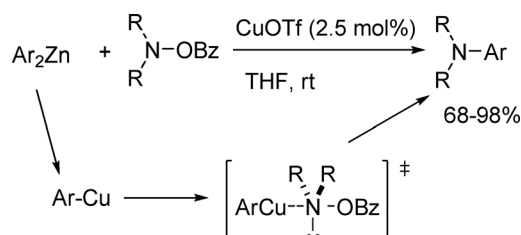
In 2004, Johnson and coworkers reported their pioneering work on the copper-catalyzed electrophilic amination of diorganozinc reagents.¹⁶ *O*-Benzoylhydroxylamines were employed in this reaction, affording tertiary amines with hindered substituents in high yields catalyzed by CuOTf (Scheme 9). It is noteworthy that it is convenient both in terms of reaction conditions (room temperature, ≤1 h) and ease of product isolation (acid/base extractive workup). This reaction also exhibited great functional group tolerance such as CN, CO₂R, NO₂, OTf, X, *etc.*¹⁷ The mechanism of the process was also investigated and subsequent transmetalation and S_N2 reaction on the nitrogen atom were suggested.¹¹

Caution: Copper-catalyzed amination of diorganozincs is an exothermic reaction. The released heat hardly affects the reaction yield on a micro-scale. However, when the reaction is performed on a large-scale, the yield will decrease significantly if the reaction temperature is not kept at 5 °C.¹⁸

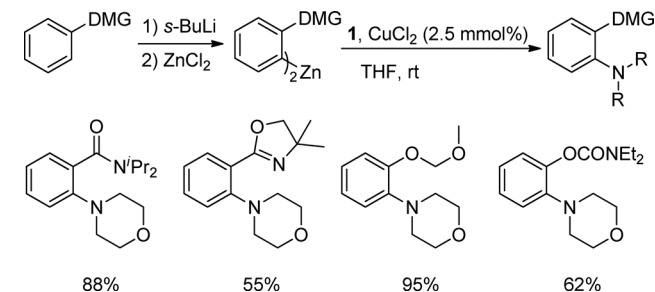
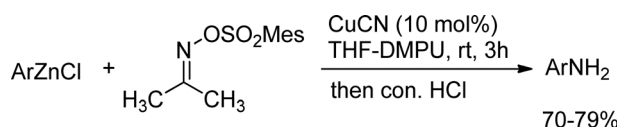
Directed *ortho* lithiation was a reliable tool for the functionalization of poorly reactive C_{aryl}–H bonds. Combined with directed *ortho* metalation/transmetalation/catalytic amination, the *ortho*-amination of the C–H reaction was achieved in one pot, affording the functionalized anilines in high yields (Scheme 10).⁶

CuCN-catalyzed amination of organozinc chlorides and diorganozincs with *O*-sulfonyloxime at room temperature gives primary amines in low to moderate yields. The addition of a dipolar aprotic solvent, as co-solvent, leads to increased yields (Scheme 11),¹⁹ while the addition of a P-, N-, or S-donor ligand would significantly decrease the quantity of copper catalyst required as well as remarkably increase the yield.²⁰

Recently, Wang's group reported the direct amination of heteroarenes and arenes through a one-pot C–H zincation and copper-catalyzed electrophilic amination procedure. This



Scheme 9 CuOTf-catalyzed amination of diorganozinc reagents.

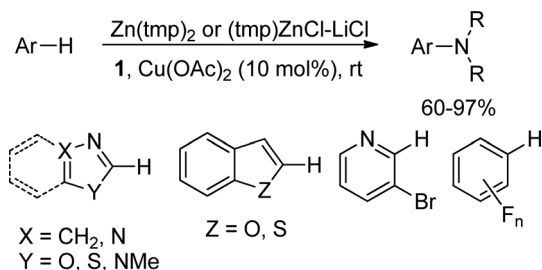
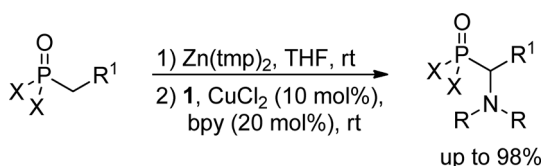
Scheme 10 Directed *ortho* metalation/amination reactions.

Scheme 11 CuCN-catalyzed amination of organozinc reagents.

procedure provided an efficient and rapid approach to access a diverse range of heteroaromatic and aromatic amines (Scheme 12). Its mild reaction conditions and good functional-group compatibility demonstrates its potential for the synthesis of important and complex amines.²¹

This C–H zincation/electrophilic amination protocol can also be applied to acidic sp^3 C–H bonds. Wang and coworkers also reported a zincation/electrophilic amination approach to α -amino phosphonic acids and their derivatives (Scheme 13). This α -amination strategy enabled the direct introduction of a variety of acyclic and cyclic amines to the α -position of phosphonates and phosphine oxides for the first time.²²

Under similar conditions, α -amination reactions of esters and amides were achieved *via* a copper-catalyzed electrophilic amination of their zinc enolates using *O*-acylhydroxylamines. It is worth mentioning that, in the absence of a copper catalyst,

Scheme 12 Cu(OAc)₂-catalyzed amination of heteroarenes and arenes.Scheme 13 CuCl₂-catalyzed amination of phosphonates and phosphine oxides.

the direct treatment of zinc enolates with *O*-acylhydroxylamines would form α -acylated 1,3-dicarbonyl products exclusively (Scheme 14).²³ This also provides insight into the dual electrophilic reactivity of *O*-acylhydroxylamines for selective amination and acylation reactions of other C–H bonds.

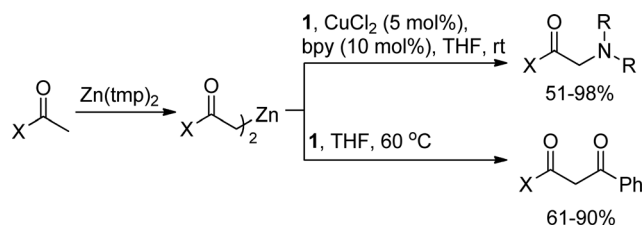
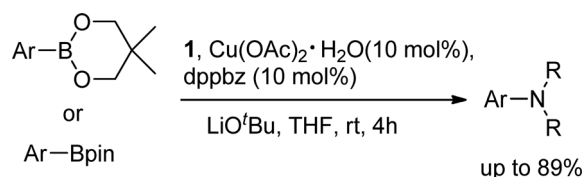
3.3 Electrophilic amination with organoboronic acids

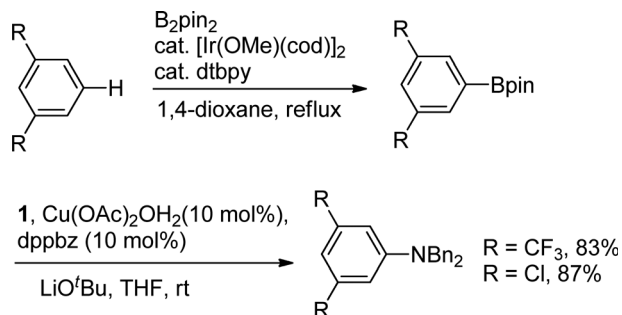
Organoboronic acids and their derivatives are used extensively in organic chemistry as building blocks or intermediates predominantly in cross-coupling reactions.²⁴ They are generally stable, readily available, and compatible with a wide range of functional groups. In 1998, the groups of Chan, Lam, and Evans independently developed Cu-mediated oxidative coupling reactions between nucleophilic arylboronic acids/esters and amines.⁴ This reaction, named Chan–Lam coupling, provided a complementary method for the formation of aniline derivatives. However, this method can only be applied to a limited scope of amine substrates.

In 2012, Miura and coworkers reported a Cu(OAc)₂·H₂O-catalyzed electrophilic amination of arylboronates with *O*-benzoylhydroxylamines (Scheme 15).²⁵ This umpolung amination approach enables the use of secondary acyclic amines, which are substrates with relatively low reactivity for the conventional Chan–Lam coupling. Moreover, the catalysis accommodates a diverse set of functional groups including chlorides, bromides and iodides, as well as aldehydes, ketones, and esters.

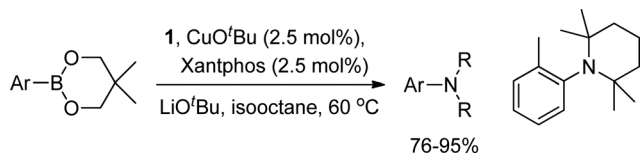
Moreover, the Cu(OAc)₂·H₂O-catalyzed electrophilic amination of arylboronates can be easily combined with the iridium-catalyzed direct C–H borylation²⁶ of simple arenes. For example, 1,3-disubstituted benzenes were converted to the corresponding anilines in one pot as shown in Scheme 16.

Lalic's group also reported a Cu^tBu-catalyzed electrophilic amination reaction for the synthesis of hindered anilines from aryl and heteroaryl boronic esters, which is compatible with a wide variety of functional groups (Scheme 17).²⁷

Scheme 14 Selective α -amination and α -acylation of esters and amides.Scheme 15 Cu(OAc)₂-catalyzed amination of arylboronates.



Scheme 16 C–H borylation/electrophilic amination sequence.

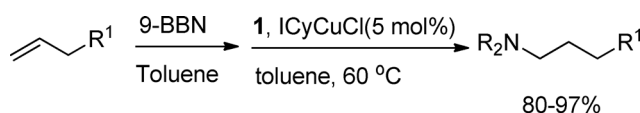


Scheme 17 CuOtBu-catalyzed amination of arylboronates.

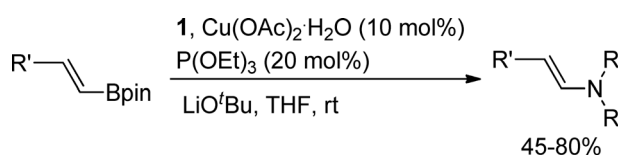
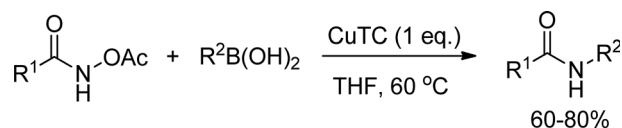
Later on, Lalic and coworkers reported a ICyCuCl-catalyzed amination of alkyl boranes through hydroboration of alkenes.²⁸ This one-pot procedure involves the hydroboration of an alkene followed by a novel electrophilic amination of the alkyl borane catalyzed by an NHC–Cu complex. Terminal alkenes are successfully transformed into tertiary alkyl amines bearing a variety of functional groups. It also provides a novel procedure for the anti-Markovnikov hydroamination of alkenes (Scheme 18).

The amination of alkenylboranes was also achieved in a Cu(OAc)₂·H₂O/P(OEt)₃ catalyzed system (Scheme 19).²⁹ Furthermore, the sequential Zr/Cu catalysis for formal anti-Markovnikov hydroamination of terminal alkynes afforded enamines in one-pot.

The electrophilic amidation of aryl, heteroaryl, and alkenyl boronic acid with *O*-acetyl hydroxamic acids was reported by Liebeskind's group (Scheme 20).³⁰ A stoichiometric amount of CuTC was required for this procedure. Nonetheless, it allows the easy preparation of a variety of amides and is a useful complement to existing metal-catalyzed methods for C–N bond formation.



Scheme 18 Hydroamination of alkenes.

Scheme 19 Cu(OAc)₂-catalyzed amination of alkenylboronates.

Scheme 20 CuTC-mediated electrophilic amidation of boronic acid.

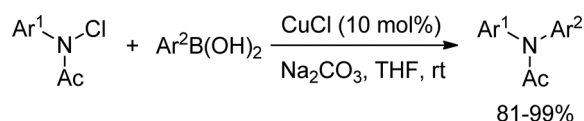
Utilizing an easily prepared and highly reactive *N*-chloroamide compound, Lei and coworkers developed an efficient copper-catalyzed electrophilic amination of arylboronic acids.³¹ The transformation gave high yields of diarylamides and tolerated a wide variety of functional groups, including iodo, bromo, and chloro moieties, which are typically sensitive in palladium-catalyzed reactions (Scheme 21).

A copper-catalyzed *N*-imination of boronic acids was also developed.³² The reaction used *O*-acyl ketoximes as the iminating agent and proceeded under non-basic and non-oxidizing conditions, thus complementing the existing C–N bond forming reactions (Scheme 22). This method tolerates various functional groups and takes place efficiently using aryl, heteroaryl, and alkenyl boronic acids.

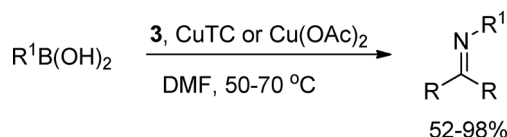
Liebeskind and coworkers further developed a simple and modular method to get highly substituted pyridines.³³ It is a cascade reaction comprising: (1) a novel *N*-iminative, Cu-catalyzed cross-coupling of alkenylboronic acids at the N–O bond of α,β -unsaturated ketoxime *O*-pentafluorobenzoates, (2) electrocyclization of the resulting 3-azatriene, and (3) air oxidation, affording highly substituted pyridines in moderate to excellent isolated yields (Scheme 23).

3.4 Electrophilic amination with organosilicon reagents

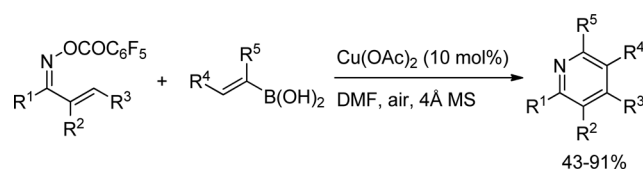
It was reported that the oxidative coupling reaction of organosilicon reagents with a variety of N–H containing



Scheme 21 CuCl-catalyzed electrophilic amidation of boronic acid.



Scheme 22 CuTC-catalyzed imination of boronic acids.

Scheme 23 Cu(OAc)₂-catalyzed synthesis of pyridines by imination/cyclization.

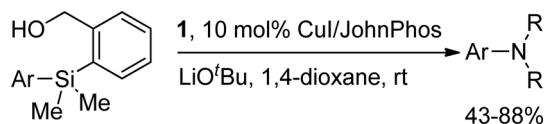
substrates afforded amines and derivatives. However, either stoichiometric amounts of copper salts³⁴ or tetrabutylammonium fluoride (TBAF)³⁵ as an additive were required. Miura and coworkers developed a copper-catalyzed electrophilic amination of aryl[(2-hydroxymethyl)phenyl]-dimethylsilanes with *O*-acylated hydroxylamines (Scheme 24).³⁶ This strategy allows arylsilanes to serve as an efficient aryl source for the construction of aniline derivatives. Moreover, the high halogen compatibility of the process provides facile access to bromide-substituted arylamines and complements the conventional Pd-based amination chemistry.

Amos and co-workers developed a copper-catalyzed electrophilic amination of aryl and heteroaryl organolithiums with *O*-benzoylhydroxylamines, using recoverable siloxane as the transfer agent (Scheme 25).³⁷ It involves a reaction of organolithiums with siloxane and an electrophilic amination of the *in situ* prepared silanes. The use of siloxane transfer agents offers a viable solution for the direct application of organolithiums in electrophilic amination.

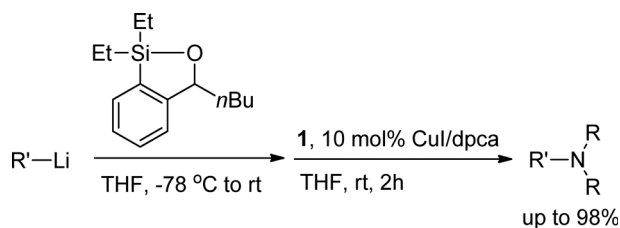
3.5 Electrophilic amination with silyl ketene acetals

α -Amination of carbonyl compounds has drawn much attention because α -amino acids and their derivatives are fundamental and important structural motifs in many biologically active compounds and pharmaceutical targets, especially peptide drugs. The traditional α -halogenation of carbonyls and subsequent nucleophilic substitution with amines is practical, but often suffers from low efficiency and fierce conditions. To overcome these limitations, electrophilic aminations of carbonyl compounds and derivatives with different amination reagents have been explored.

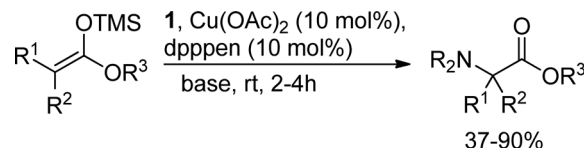
Silyl ketene acetals are commonly used as nucleophilic reagents in aldol, Michael and Mannich reactions. Miura *et al.* developed a copper-catalyzed amination of ketene silyl acetals with *O*-benzoylhydroxylamines, directing to the synthesis of unnatural α -amino acids (Scheme 26).³⁸ It provides a brand new C–N bond-forming approach to α -amino acids starting from substrates at the carboxylic acid oxidation level.



Scheme 24 CuI-catalyzed amination of organosilicon reagents.



Scheme 25 Electrophilic amination of organolithiums employing a siloxane transfer agent.



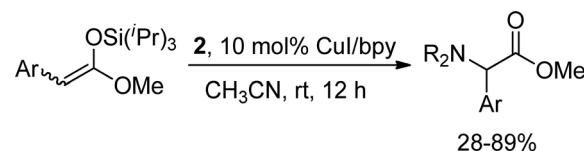
Scheme 26 $\text{Cu}(\text{OAc})_2$ -catalyzed amination of silyl ketene acetals.

The amination of silyl ketene acetals with *N*-chloroamines was also investigated by Miura and coworkers.³⁹ Using CuI/bpy as the catalyst, α -amino acids were obtained with high yields under mild conditions (Scheme 27). The facile availability of *N*-chloroamines from secondary amines permits the one-pot two-step synthesis of α -amino acids starting from amines on a gram scale.

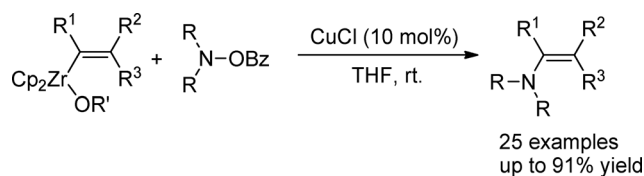
3.6 Electrophilic amination with organozirconium reagents

So far, most of the electrophilic amination reactions have been achieved with main-group organometallic reagents. Jarvo and coworkers reported a titanium-mediated amination of Grignard reagents with *N*-chloroamines, which was generated *in situ* from amines and NCS.⁴⁰ Aryltitanium, produced *in situ* by reacting ArMgX with $\text{Ti}(\text{O}^i\text{Pr})_4$, is a suitable nucleophilic reagent to react with *N*-chloroamines, forming amines in good yields. As part of an ongoing program in our laboratory to systematically study zirconocene chemistry, we reported a copper-catalyzed amination of alkenylzirconocenes with *O*-benzoylhydroxylamines, affording a wide range of enamines under mild conditions (Scheme 28).⁴¹ The double bond was retained during the whole process. This reaction also tolerates a wide range of functional groups and can be used for the preparation of hindered enamines. Mechanistically, we proposed that the alkenylzirconocene undergoes transmetalation with CuCl before amination.

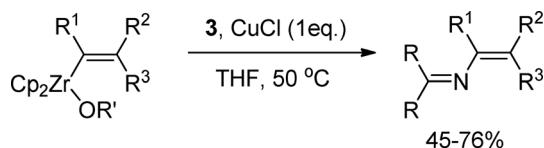
When *O*-benzoyloximes were utilized, electrophilic imination with alkenylzirconocenes afforded 2-azadienes in the presence of a stoichiometric amount of CuCl (Scheme 29).⁴² Both *O*-benzoylketoximes and *O*-benzoylaldoximes can be employed in this reaction. The configurations of the C=C and C=N double bonds persisted during the imination.



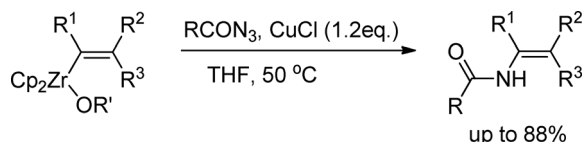
Scheme 27 CuI-catalyzed amination of silyl ketene acetals.



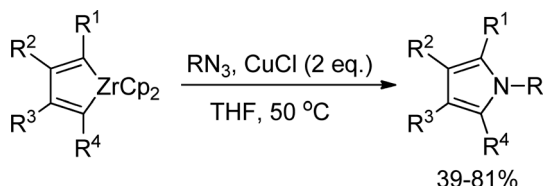
Scheme 28 CuCl-catalyzed amination of alkenylzirconocenes.



Scheme 29 CuCl-mediated imination of alkenylzirconocenes.



Scheme 30 CuCl-mediated amidation of alkenylzirconocenes.



Scheme 31 Reaction of zirconacyclopentadienes with azides.

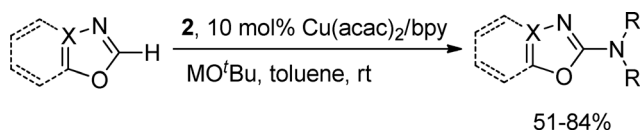
The amidation reaction of alkenylzirconocenes with acyl azides was also investigated (Scheme 30).⁴³ This reaction can be used to prepare various multi-substituted enamides.

Zirconacyclopentadienes were considered as two alkenylzirconocenes. When they were treated with azides, pyrroles formed *via* double amination (Scheme 31).⁴⁴ This offers a new approach to form multi-substituted pyrroles through the Zr/Cu mediated cyclization of two alkynes and one azide in one pot.

4. Electrophilic amination of direct C–H bonds

Complementary to the electrophilic amination with organo-metallic reagents, direct C–H amination with electrophilic aminating reagents provides a new and potentially more effective C–N bond formation approach, although several effective metal-catalyzed C–H/N–H coupling methods were achieved *via* oxidative coupling.³ The first direct C–H amination was reported by the group of Miura, which is a Cu-catalyzed reaction of *N*-chloroamines and azoles (Scheme 32).⁴⁵ This reaction proceeded under rather mild conditions and gave straightforward access to the heteroarylamines.

Miura's group also investigated the C–H amination of polyfluoroarenes and azoles with *O*-benzoylhydroxylamines



Scheme 32 Cu(acac)₂-catalyzed direct C–H amination of azoles.

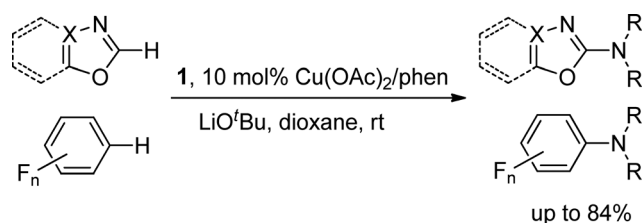
(Scheme 33).⁴⁶ The catalysis enabled the rapid and concise construction of polyfluoroanilines and aminoazoles, which are vital in pharmaceutical and medicinal chemistry.

As early as 2002, Narasaka and coworkers reported the intramolecular amination of oxime derivatives. The radical cyclization of γ,δ -unsaturated *O*-methoxycarbonyl or *O*-pentafluorobenzoyloxime afforded 3,4-dihydro-2*H*-pyrroles using CuBr·Me₂S as the catalyst.⁴⁷ They also discovered the cyclization of β -(3-indolyl) ketone *O*-pentafluorobenzoyloximes to generate 3,4-dihydro- α -carboline (Scheme 34).⁴⁸

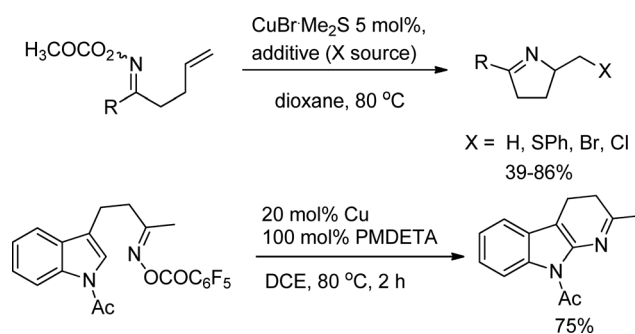
Bower and coworkers recently reported the intramolecular amination of an allylic C–H bond. The copper-catalyzed azo-Heck cyclization of oxime esters afforded pyrrolidine derivatives (Scheme 35).⁴⁹ Mechanistic studies indicated a reaction pathway proceeding *via* the generation and cyclization of an intermediate that possessed iminyl radical character.

Guan and coworkers reported a copper-catalyzed 5-*endo*-trig cyclization of β,γ -unsaturated ketoxime carboxylates for the synthesis of 2-arylpyrroles (Scheme 36).⁵⁰ This reaction tolerates a range of functional groups and is a practical procedure for rapid synthesis of 2-arylpyrroles.

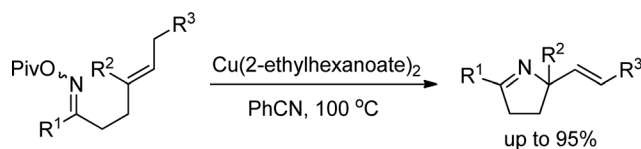
The intramolecular amination of amidoximes was recently investigated by Chiba's group. CuI-catalyzed reactions of



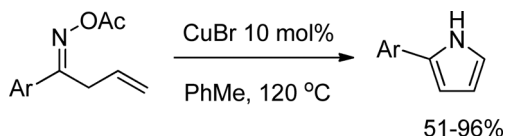
Scheme 33 Cu(OAc)₂-catalyzed direct C–H amination of azoles and polyfluoroarenes.



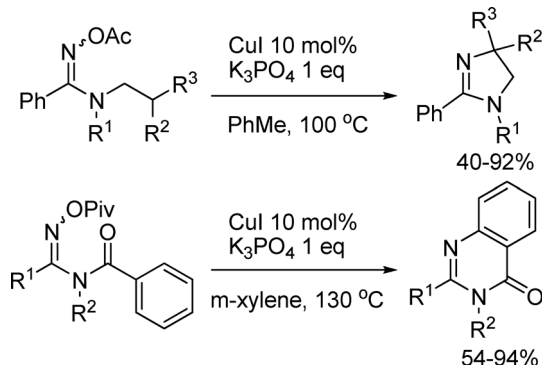
Scheme 34 Copper-catalyzed intramolecular amination of γ,δ -unsaturated oxime esters.



Scheme 35 Cu(2-ethylhexanoate)₂-catalyzed intramolecular amination of γ,δ -unsaturated oxime esters.



Scheme 36 CuBr-catalyzed intramolecular amination of β,γ -unsaturated oxime esters.

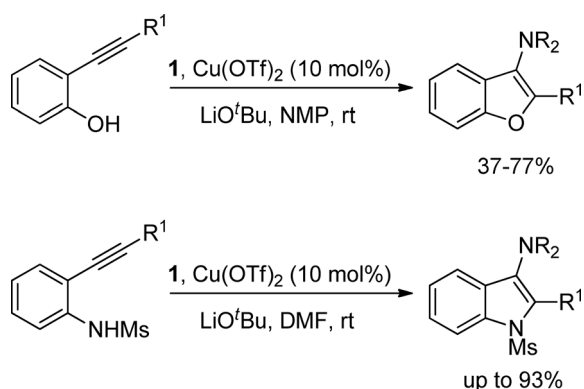


Scheme 37 CuI-catalyzed intramolecular amination of amidoximes.

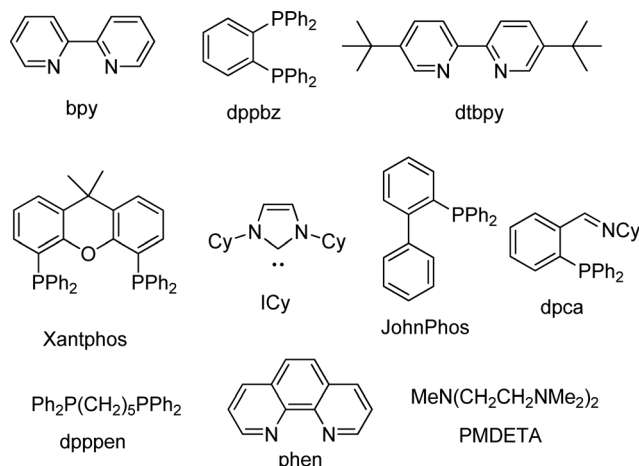
N-alkylamidoximes afforded dihydroimidazoles *via* an sp^3 C–H amination. The reactions of *N*-benzoylamidoximes resulted in sp^2 C–H amination forming quinazolinones (Scheme 37).⁵¹

5. Annulative electrophilic amination of *o*-alkynylphenols and *o*-alkynylanilines

The metal-catalyzed electrophilic annulation of *o*-alkynylphenols and anilines was an efficient approach to construct functionalized benzofurans and indoles. Muira and coworkers developed an effective copper-catalyzed annulative amination of *ortho*-alkynylphenols with *O*-benzoylhydroxylamines. The system may compensate for the precedents and provide a facile new route for the 3-aminobenzofuran skeletons which are of biological and pharmaceutical interest.⁵² The annulative amination of *o*-alkynylanilines afforded the corresponding 3-aminoindoles (Scheme 38).⁵³



Scheme 38 Cu(OTf)₂-catalyzed annulative amination of *o*-alkynylphenols and *o*-alkynylanilines.



Scheme 39 Structures and abbreviations of ligands used in this perspective.

6. Conclusions and outlook

Copper-mediated electrophilic amination of organometallic reagents is an important and useful synthetic method for the preparation of aliphatic, aromatic, and heteroaromatic amines, as well as enamines and enamides and is continuing to gain the attention of chemists. Furthermore, copper-catalyzed direct C–H amination with electrophilic aminating reagents provides a new and potentially more effective C–N bond formation approach for the synthesis of aromatic and heteroaromatic amines. We hope this perspective has reflected the rapid development of interest in this field. Further work is needed to extend the scope of the organometallic reagents and direct C–H reagents, as well as the electrophilic aminating reagents.

Abbreviations

Abbreviations of the ligands involved in this perspective are shown in Scheme 39.

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